

Documentation gaps and verification queue

Status: source-reviewed, partial browser hardware check (firmware snapshot: 2026-09-23). The earlier explicit-sync failure was reproduced and resolved on the assembled boards; other entries still need hardware verification. Resolve or confirm them before turning the development guides into a public manual.

Priority	Finding and source	Documentation consequence / next check
Medium	The reproduced explicit-sync timeout was resolved by parsing the Uno's fixed six-byte status replies directly on the Pico. The automated release-build sequence stayed connected through explicit sync, Randomize, and patch loads. Pico link parser , Uno receive loop , test record .	Remaining acceptance: measure the Pico RX high voltage, disconnect the return wire to check offline detection, force a lost acknowledgment, and test both boards starting together and either board starting late.
High	Standalone activity is tracked when DIN or RTP-MIDI delivers Note On/Off, CC, pitch bend, or an accepted QuarkWave sound command; MIDI clock and transport are excluded. No physical-controller-presence detection exists. Uno external handlers , Pico auto-sync rule .	Run the Uno without the Pico, attach a controller, change the sound, then attach the Pico. Confirm the sound persists until explicit Sync Pico patch to Uno , and that the warning precedes the reset.
Medium	Fresh user slots are now empty, while any existing file—including a legacy <code>P0 ... P7</code> file or unreadable file—is preserved as occupied. Save As requires an explicit destination and confirmation for replacement. Pico storage and save handler , Save As dialog .	On hardware, check blank storage, upgrades with legacy files, full slots, unreadable files, write failures, and reboot after loading a factory or user patch. Confirm save-result messages and displayed selection.
Medium	The redesigned panel wires its visible buttons to the existing <code>type</code> messages; Randomize sends one <code>randomize</code> request to the Pico and returns <code>patchData</code> . Browser rendering and control messages were checked in headless Chrome; per-control Uno MIDI reception and audio remain unverified. Browser handlers and view switching , Pico dispatch .	Browser checks passed for 50 control paths, 41 pointer keys, view switching with a held browser note, Randomize, Load, Save As dialog, rejected saves, visualization messages, and Panic message. Test Uno response and audio, successful Save/Save As, Commit/Load Commit, and repeat view/keyboard checks after link recovery.
Medium	DIN and RTP now accept sound Program Changes and QuarkWave SysEx 01–03, while visualization, reset, and link commands remain Pico-only. Each input has its own clock history, with DIN priority and transport-controlled arpeggiator scheduling. Uno input handlers .	On hardware, test accepted and rejected commands, 40/120/240 BPM clock, beat alignment, Start/Continue/Stop, DIN priority, two-second failover, and held tempo after clock loss.

Priority	Finding and source	Documentation consequence / next check
Medium	The owner's photographs show the two USB-powered boards and level-shifter module; the owner confirms the return-wire shift and common ground. Both UART signal wires appear to use separate level-shifter channels; exact channel pairs, LV/HV supply terminals, and Pico RX voltage are unmeasured. The Uno's <code>AO</code> audio stage is not built. The owner also has a TDA1308 headphone board , not yet connected; DIN hardware is not shown. Pico wiring comment , Uno MIDI instance , DAC write .	Use the current inter-board schematic to continuity-check both UART paths, supply terminals, and common ground; measure Pico RX. Build and test the proposed mono line and headphone outputs before treating their diagrams as as-built or quoting audio levels. Check the headphone board's ground and output DC before listening. Document any DIN interface separately.

Use the [hardware test log](#) to record observations for each assembled revision.

The older-panel deployment gap was resolved on 2026-09-23: the current panel loaded after flashing the Pico W, and its Pico/Uno labels reached **Connected** / **Connected · Pico patch**. Automated browser checks cover controls and keys. The explicit-sync link failure has been reproduced, fixed, and retested on both boards; audio and per-command Uno reception checks remain open.

Hands-on acceptance pass, when both boards are available

1. Record board/core and library versions, the three source hashes from the [index](#), and the assembled wiring revision.
2. Record the owner-confirmed return-wire level shift, continuity-checked jumper endpoints, and measured Pico RX voltage, then power both boards via their USB ports; check Wi-Fi, `quarkwave.local`, IP fallback, separate **Pico** and **Uno** connection labels, note playback once the audio stage exists, and note release on browser disconnect.
3. Compare browser controls with Uno behavior for each group: envelope, tone, modulation, arpeggiator, delay, chorus, bitcrush, tremolo, drive/fold, and Panic. Log MIDI CC values and compare panel readouts, including vibrato rate and master volume; repeat Randomize and check its full-range effects. Check keyboard velocity 1, 100, and 127 and view switching with a held note.
4. Save and load a user patch, factory preset, factory-to-user copy, and Commit snapshot; restart each board separately and confirm the selected Pico patch is restored on clean Uno startup without opening a browser. Test a lost completion acknowledgment and return-wire disconnection.
5. Exercise the separate DIN and RTP paths with notes, bend, CC, and MIDI clock, both with and without the Pico attached. Confirm standalone activity blocks automatic takeover, while clock alone does not. Check the documented Program Change and SysEx allowlist on each input, including malformed messages and standalone status. Check that BLE is absent in this build.
6. Update the guides with verified results, then remove resolved entries from this queue or link them to development issues.